12TH BIOLOGY · CHAPTER 15 · CHAPTER TEST

Human Nervous System

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 15	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Neurotransmitter receptors are located on:

(a) Nuclear membrane (b) Nodes of Ranvier
(c) Myelin sheath (d) Postsynaptic membrane

2. The resting membrane potential of a neuron is about:

(a) +50 mV (b) –70 mV
(c) 0 mV (d) –120 mV

3. Which part of the nervous system prepares the body during emergency?

(a) Sympathetic nervous system (b) Central nervous system
(c) Parasympathetic nervous system (d) Sensory pathway

4. Dorsal root of spinal nerve contains:

(a) Motor neurons (b) Inter-neurons
(c) Sensory neurons (d) Sensory and motor neurons

5. Which receptor detects deep pressure and vibration?

(a) Meissner’s corpuscles (b) Pacinian corpuscles
(c) Merkel cells (d) Ruffini endings

6. Caffeine increases alertness by:

(a) Increasing dopamine receptors (b) Stimulating nociceptors
(c) Blocking adenosine receptors (d) Enhancing serotonin degradation

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. State the path of a message transmitted through the nervous system.
ii. Name the factors responsible for the resting membrane potential of neuron.
iii. Differentiate between white matter and grey matter.
iv. What is the refractory period?
v. Define narcotic drugs and give examples.
vi. State the causes and symptoms of Alzheimer’s disease.

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain synaptic transmission of nerve impulse.
(b) Explain the types and functions of autonomic nervous system.

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